

Contents

Capabilities Summary 1

Gartner MQ / Critical Capabilities — AI Application Development Platforms 2026 1

1. Deep research agent 1

1.1 The agent itself 1

1.2 The lifecycle around it 1

2. External customer AI assistant 2

3. Customer agent — grounding, guardrails, DLP, reporting 2

4. Voice experience and the avatar 2

5. Services used to build the solution 3

Agent platform 3

Models 3

Data and knowledge 3

Build and delivery 3

Third-party components, disclosed 3

6. Declared exceptions 4

Capabilities Summary

Gartner MQ / Critical Capabilities — AI Application Development Platforms 2026

What Gartner asked us to demonstrate, and the AWS service that answers each item. Organised in the four sections of Gartner's brief.

Companion documents: `demo-script.docx` for what appears on screen section by section, and `capabilities-mapping.docx` for the full requirement-level detail.

1. Deep research agent

Gartner asks for the whole development lifecycle here, not just a finished agent.

1.1 The agent itself

Gartner asks for	AWS service
Build with an agentic framework	Strands Agents SDK on Amazon Bedrock AgentCore Runtime
Take user input, refine and optimise the query with an LLM	Claude Sonnet 5 on Amazon Bedrock
Break the query into sub-questions	Planner agent, Claude Sonnet 5
Research plan naming objectives, methods, evaluation criteria, expected outcomes	Planner agent emitting structured JSON
Generate search queries from the refined question	Researcher agent, Claude Sonnet 5
Human in the loop to review and refine the plan	Two-phase plan/approve flow; AgentCore Gateway elicitation
Tool calling / MCP, such as web search APIs	AgentCore Gateway (MCP), Amazon Nova Web Grounding
Divide research into sections; multi-agent collaboration for parallel processing	AgentCore Runtime, multi-agent supervisor pattern
Human in the loop to refine content	AgentCore Gateway elicitation
Multimodal input, e.g. data-analysis charts	Claude vision on Bedrock; AgentCore Code Interpreter
Multimodal output — graphics, logos, storefronts	Amazon Nova Canvas (images), Amazon Nova Reel (video)
One chapter as text, images and speech	Nova Canvas, Amazon Nova Sonic 2
Combine sections with introduction and conclusion	Synthesizer agent, Claude Sonnet 5
Citations for all sources	Nova Web Grounding citations carried through to the runtime
Export as PDF	Lambda PDF generator, Amazon S3

1.2 The lifecycle around it

Gartner asks for	AWS service
IDE-based development	Kiro — specs (requirements, design, tasks) and steering files
Low-code development	Amazon Bedrock AgentCore Harness — the agent as configuration
High-code development	Strands Agents SDK; export from harness to code
Graphical development	AgentCore console; AWS Step Functions visual builder
Model catalog	Amazon Bedrock model catalog and cross-Region inference profiles
Testing and validation	Kiro-generated tests; AgentCore Evaluations
Data preparation	Amazon Bedrock Data Automation; Managed Knowledge Base ingestion

Gartner asks for	AWS service
Tracing	AgentCore Observability; AWS X-Ray
Evaluation	AgentCore Evaluations — online, batch, user simulation
Debugging	AgentCore Observability span trees; Agent Inspector
Agent operations	Amazon CloudWatch dashboards; AgentCore Observability
Orchestration	AgentCore Runtime; AgentCore Gateway; AWS Step Functions
Cost management	CloudWatch cost attribution; AWS Budgets; resource tagging
Control processes	AgentCore Policy (Cedar); Amazon Bedrock Guardrails; Kiro steering; evaluation release gates
Deployment	AWS CDK; AgentCore CLI; Amazon ECR; AWS CodeBuild; Amazon EKS and Helm for hybrid

2. External customer AI assistant

Gartner asks for	AWS service
Assistant grounded on the section 1 output	Amazon Bedrock Managed Knowledge Base over the generated PDF and
Detailed service information as text and images	Managed Knowledge Base multimodal retrieval; Bedrock Data Automati
Generate service images	Amazon Nova Canvas
Speak account and investment progress	Amazon Nova Sonic 2 speech-to-speech
Automatic quality control — format, length, filtering, irrelevance	Response schema validation; Bedrock Guardrails; AgentCore Evaluations
Human review to update service descriptions	AgentCore Gateway elicitation; versioned content in Amazon S3
A/B test output across models	AgentCore Optimization — configuration bundles and gateway traffic sp
Continuous feedback loop	AgentCore Observability, Evaluations, Optimization recommendations

3. Customer agent — grounding, guardrails, DLP, reporting

Gartner asks for	AWS service
Open an account	AgentCore Runtime with AgentCore Gatew
Validate the bank can do business with the customer	Eligibility and sanctions tools as Gateway t
Let customers take up services	Gateway tools writing to Amazon Dynamo
Detect fraudulent accounts and transactions	Second agent via agent-to-agent call; Agent
Grounding by context and prompt engineering	Structured context assembly in the agent, v
Grounding by RAG	Amazon Bedrock Managed Knowledge Base
Quickly verify answers are grounded in the section 1 document	Citations with source and page, checked ag
Guardrails block requests unrelated to the bank	Amazon Bedrock Guardrails — denied topi
Employee-salary questions trigger guardrails	Bedrock Guardrails; entitlement-aware retr
Data loss prevention	Bedrock Guardrails sensitive-information fi
Out-of-the-box reports — comprehensiveness, accuracy, response time, cost, marketing metrics	AgentCore Observability and Evaluations d

4. Voice experience and the avatar

Gartner asks for	AWS service
Photorealistic avatar	Partner rendering layer on the AWS voice stream — disclosed, not demonstrated
At least two languages	Amazon Nova Sonic 2, native multilingual
Guidance in multiple languages and voice tones	Nova Sonic 2 voice and prosody selection
Ground the avatar to the section 2 text and images	The same Bedrock Managed Knowledge Base as section 2
Spoken question and answer on why to choose the bank	Nova Sonic 2 with AgentCore Gateway tools
Show or indicate the item being discussed	Agent-emitted UI tool call through AgentCore Gateway
Change item when the customer asks	Same UI tool, re-invoked mid-conversation
Test guardrails with out-of-scope questions	Amazon Bedrock Guardrails, the same rule set as the text channels

5. Services used to build the solution

Agent platform

Service	Role
Amazon Bedrock AgentCore Runtime	Hosts the agents; session isolation, streaming, long-running jobs
Amazon Bedrock AgentCore Gateway	Publishes every tool as MCP; web search; elicitation
Amazon Bedrock AgentCore Memory	Preferences, session summaries, recall across runs
Amazon Bedrock AgentCore Identity	Agents act as the signed-in user; on-behalf-of delegation
Amazon Bedrock AgentCore Policy	Deterministic allow and deny ahead of the model (Cedar)
Amazon Bedrock AgentCore Observability	Traces, spans, latency, token and cost attribution
Amazon Bedrock AgentCore Evaluations	Quality gates; online and batch scoring
Amazon Bedrock AgentCore Optimization	Prompt recommendations, configuration bundles, A/B tests
Amazon Bedrock AgentCore Browser	Sources with no API, with human takeover
Amazon Bedrock AgentCore Code Interpreter	Sandboxed analysis and chart generation
Amazon Bedrock AgentCore Harness	The agent declared as configuration

Models

Service	Role
Claude Sonnet 5, Opus 5, Opus 4.7, Sonnet 4.6, Haiku 4.5	Reasoning, planning, synthesis; selectable per session
Amazon Nova Sonic 2	Real-time speech-to-speech
Amazon Nova 2 Lite	Low-cost tool selection and web grounding
Amazon Nova Canvas	Image generation and editing
Amazon Nova Reel	Video generation
Amazon Nova 2 Multimodal Embeddings	Knowledge base embeddings

Data and knowledge

Service	Role
Amazon Bedrock Managed Knowledge Base	Managed RAG pipeline with native connectors
Amazon S3 Vectors	Vector store for the knowledge base
Amazon Bedrock Data Automation	Parses documents and imagery for ingestion
Amazon Bedrock Guardrails	Content, topic and sensitive-information policy
Amazon S3	Documents, generated PDFs, images, website assets
Amazon DynamoDB	Application records and session metadata

Build and delivery

Service	Role
Kiro	Spec-driven design, steering as governance-as-code, generated tests
AWS CDK (TypeScript)	All infrastructure as code
AgentCore CLI	Scaffold, run locally, deploy, invoke
AWS CloudFormation	Deployment engine behind CDK
Amazon ECR, AWS CodeBuild	Container images for the agent runtimes
Amazon ECS on AWS Fargate	The real-time voice worker (ARM64)
AWS Lambda	Gateway tool implementations
Amazon API Gateway	Token and feedback endpoints
Amazon Cognito	Authentication and per-user identity
AWS Secrets Manager, AWS Systems Manager Parameter Store	Credentials and runtime configuration
Amazon CloudWatch, AWS X-Ray	Logs, metrics, dashboards, tracing
AWS WAF, AWS KMS	Edge protection and encryption
Amazon CloudFront, Amazon S3	Frontend hosting

Third-party components, disclosed

Component	Licence	Role
LiveKit Cloud	Commercial	WebRTC transport for the real-time voice channel
Strands Agents SDK	Apache-2.0	Agent framework for the code-level implementation
TalkingHead, HeadAudio	MIT	Browser avatar rendering and audio-to-viseme lip sync
Avatar rendering partner	Commercial	Photorealistic avatar layer — disclosed, not demonstrated

6. Declared exceptions

Stated plainly because Gartner does not evaluate capabilities that are not generally available, and the source code is a deliverable.

Item	Position
AgentCore Payments	Public preview. Not demonstrated, declared as beta.
AgentCore Failure Insights	Public preview. Not demonstrated. Recommendations, batch evaluation and A/B testing are GA and a
AWS Agent Registry	Generally available 6 August 2026. Gateway connectors, the AWS-curated skills catalog and AWS Mar
Photorealistic avatar rendering	Partner technology. AWS supplies speech, grounding, tool calls and guardrails; the rendering layer is d
Terraform for AgentCore CLI	Not yet available. AWS CDK is used and claimed.
Claude Agent SDK harness export	Not yet available. Strands export is used and claimed.